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Classification of Meristem

Position	Origin	Function	Plane of division
Apical meristem Present in apices of root and shoot. It is responsible for increase in the length of the plant, it is called as primary growth.	**Primary Meristem** It is derived from embryonic stages and differentiated into primary permanent tissues.	**Protoderm** It gives rise to epidermal tissue system and develops into epidermis,stomata and hairs.	**Mass meristem** It divides in all planes. Example: endosperm,young embryo and sporangium
Intercalary meristem Occurs between the mature tissues. It is responsible for elongation of internodes.	**Secondary meristem** It is derived during later stage of development of the plant body. It produces cork cambium and interfascicular cambium.	**Procambium** It gives rise to primary vascular tissues. Example: xylem and phloem .	**Rib meristem or File meristem** It divides anticlinally in one plane. Example: development of cortex and pith
Lateral meristem Occurs along the longitudinal axis of stem and root. It is responsible for secondary tissues and thickening of stem and root. Example: vascular cambium and cork cambium.		**Ground Meristem** It gives rise to all tissues except epidermis and vascular strands.	**Plate meristem** It divides anticlinally in two planes. Example: development of epidermis

Theories of Meristem Organization and Function

Many anatomists illustrated the root and shoot apical meristems on the basis ofnumberandarrangementandaccordingly proposed the following theories – An extract of which are discussed below.

Shoot Apical Meristem

Apical Cell Theory

Apical cell theory is proposed by **Hofmeister** (1852) and supported by **Nageli** (1859). A single apical cell is the structural and functional unit.

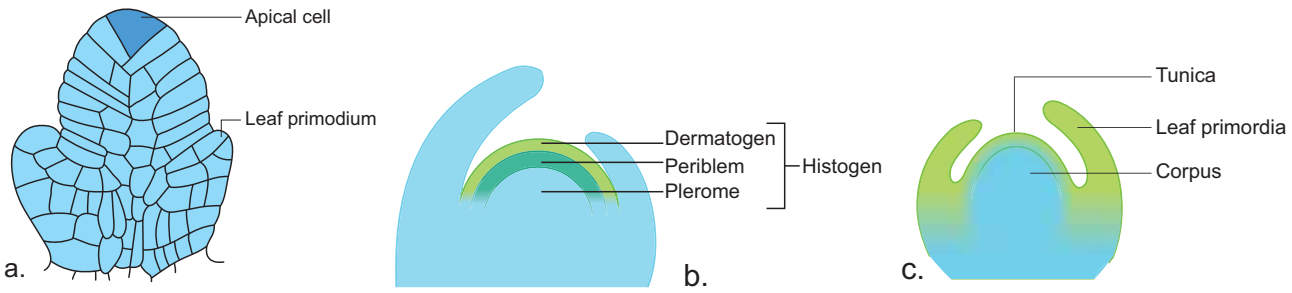


Figure 9.2: Shoot apical meristem a) Apical cell theory, b) Histogen theory, c) Shoot Tunica corpus theory